

AirCore RS-37 / RS-55

Rotary screw air compressors - Service manual

Manufacturer: SampleAir (fictional)

SAMPLE DOCUMENT. The manufacturer, models, part numbers and procedures are invented for software development. Do not use for real equipment.

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1. General safety

- Always stop the compressor and apply lockout/tagout on the main isolator before any work.
- Release all stored pressure. The separator tank stays pressurized after stop until blowdown is complete.
- Oil and internal parts can exceed 90 C. Allow at least 15 minutes to cool.
- Electrical work only by qualified electricians. Verify zero voltage before touching terminals.
- Never adjust, plug or remove the safety valve.
- Wear gloves, safety glasses and hearing protection (noise up to 72 dB(A)).

2. Technical data

	RS-37	RS-55
Motor power	37 kW	55 kW
Drive	V-belts	Direct, coupling
Free air delivery at 8 bar	6.1 m3/min	9.4 m3/min
Max working pressure	10 bar	10 bar
Safety valve setting	11 bar	11 bar
Oil capacity	22 L	30 L
Integrated dryer	No	Yes

3. Controller and emergency stop

The controller shows alarms as codes (see the alarm table in chapter 16). Shutdown alarms stop the compressor; warnings allow it to run.

Troubleshooting: Emergency stop - alarm E001

Symptom: Emergency stop alarm active; compressor will not start or reset.

Possible cause: Emergency stop button pressed, or its contact block faulty.

Procedure:

1. Find out why the e-stop was pressed and make sure it is safe.
2. Twist to release the emergency stop button.
3. Reset the alarm on the controller.
4. If the alarm stays, check the e-stop contact block with a multimeter (qualified electrician).

Parts: E-stop contact block (if faulty)

Safety: Confirm the reason for the emergency stop before resetting. Electrical work: qualified electrician only; verify zero voltage.

4. Air intake

The intake filter protects the airend from dust. Replace the element every 2000 running hours or when the differential pressure warning appears.

Troubleshooting: Air intake filter - alarm E210

Symptom: Air filter differential pressure warning; reduced air delivery; pressure builds slowly.

Possible cause: Air intake filter element clogged.

Procedure:

1. Stop the unit.
2. Open the filter housing and remove the element.
3. Replace the element (do not clean paper elements with compressed air).
4. Clean the housing and reset the service counter.

Parts: Air filter element AF-37 / AF-55

Safety: Stop the compressor and apply lockout/tagout on the main isolator.

5. Oil system

Use only SA-Synth 46. Oil and oil filter interval: 4000 running hours or 12 months.

Troubleshooting: Oil system - alarm E101

Symptom: High temperature alarm; oil level below the minimum mark in the sight glass.

Possible cause: Low oil level in the separator tank.

Procedure:

1. Stop the unit and release pressure.
2. Wait 5 minutes for oil to settle and read the sight glass.
3. Top up with the specified compressor oil to the middle of the sight glass.
4. Inspect for leaks (hoses, fittings, drain valve).
5. Restart and check temperature.

Parts: Compressor oil SA-Synth 46

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar. Hot surfaces: let the unit cool at least 15 minutes before opening.

Troubleshooting: Oil hoses and fittings

Symptom: Oil on the floor or inside the canopy; oil level dropping quickly.

Possible cause: Oil leak from a hose, fitting or the oil filter gasket.

Procedure:

1. Isolate, depressurize and cool.
2. Clean the area to find the leak source.
3. Tighten or replace the leaking hose/fitting or oil filter.
4. Top up oil.
5. Run and re-check for leaks.

Parts: Oil hose or O-ring as needed, Oil filter OF-37 (if gasket leaks)

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar. Hot surfaces: let the unit cool at least 15 minutes before opening. Clean up oil spills to prevent slips.

Troubleshooting: Compressor oil - alarm E601

Symptom: Service due warning; oil dark or smells burnt; temperature higher than usual.

Possible cause: Service interval exceeded; oil degraded and oil filter clogged.

Procedure:

1. Isolate, depressurize and let cool to about 40 C (warm oil drains better).
2. Drain the oil.
3. Replace the oil filter, air filter and oil.
4. Reset the service counter.
5. Run and check for leaks.

Parts: Compressor oil SA-Synth 46, Oil filter OF-37 / OF-55, Air filter AF-37 / AF-55

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar. Hot surfaces: let the unit cool at least 15 minutes before opening.

6. Cooling and temperature control

Normal discharge temperature is 75-95 C. Shutdown occurs at 110 C (E101).

Troubleshooting: Oil cooler - alarm E101

Symptom: High discharge temperature alarm; unit trips after running under load for 20-40 minutes.

Possible cause: Oil cooler fins blocked with dust, reducing cooling.

Procedure:

1. Stop and isolate the compressor (lockout/tagout).
2. Release pressure and let the unit cool down.
3. Blow the cooler fins clean from the fan side with compressed air, then brush.
4. Check oil level and room ventilation.
5. Restart and monitor discharge temperature under load.

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar. Hot surfaces: let the unit cool at least 15 minutes before opening. Wear gloves, safety glasses and hearing protection.

Troubleshooting: Thermostatic valve - alarm E101

Symptom: Temperature rises quickly after start; oil cooler stays cold to the touch while the unit overheats.

Possible cause: Thermostatic (thermal bypass) valve stuck closed, oil bypasses the cooler.

Procedure:

1. Isolate and depressurize the compressor.
2. Let the unit cool.
3. Remove the thermostatic valve housing.
4. Replace the thermostatic element and O-rings.
5. Restart and confirm the cooler warms up after about 10 minutes.

Parts: Thermostatic element kit TV-71

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar. Hot surfaces: let the unit cool at least 15 minutes before opening.

7. Cooling fan and drive

RS-37 units are belt driven. RS-55 units are direct driven through a flexible coupling.

Troubleshooting: Cooling fan motor - alarm E305, E101

Symptom: Fan motor overload alarm; cooling fan not turning; temperature rising.

Possible cause: Cooling fan motor failed or its overload relay tripped.

Procedure:

1. Isolate power (lockout/tagout).
2. Check the fan turns freely by hand.
3. Measure fan motor winding resistance and current.
4. Reset or replace the overload relay; replace the fan motor if windings are faulty.
5. Restart and confirm fan rotation direction.

Parts: Fan motor FM-1.1kW (if faulty)

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Electrical work: qualified electrician only; verify zero voltage.

Troubleshooting: Drive belts (RS-37 only)

Symptom: Squealing noise at start or under load; reduced air delivery; black belt dust near the guard.

Possible cause: Drive belts worn or tension too low (belt-driven RS-37).

Procedure:

1. Isolate and lock out.
2. Remove the belt guard.
3. Check belt tension and wear.
4. Replace belts as a matched set and tension to specification.
5. Refit the guard and test run.

Parts: V-belt set SPB-1600 (x3)

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Never run the unit with the belt guard removed.

8. Oil separation

Separator element replacement: 8000 running hours or when separator dP exceeds 1 bar.

Troubleshooting: Oil separator element - alarm E220

Symptom: Separator differential pressure alarm; motor current higher than normal; reduced delivery pressure.

Possible cause: Oil separator element clogged (end of service life).

Procedure:

1. Isolate, depressurize and let cool.
2. Remove the separator cover or spin-on separator.
3. Replace the separator element and gaskets.
4. Check the scavenge line orifice is clear.
5. Restart and check separator dP is below 0.6 bar.

Parts: Separator element OS-37 / OS-55, Gasket set

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar. Hot surfaces: let the unit cool at least 15 minutes before opening.

Troubleshooting: Scavenge line

Symptom: Oil carryover into the air lines; high oil consumption; oil in the condensate.

Possible cause: Scavenge line orifice or strainer blocked, oil collects in the separator and carries over.

Procedure:

1. Isolate and depressurize.
2. Remove the scavenge line and sight glass.
3. Clean the orifice and strainer.
4. Check the separator element for damage.
5. Reassemble and verify oil flow in the scavenge sight glass.

Parts: Scavenge strainer SS-10 (if damaged)

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar. Hot surfaces: let the unit cool at least 15 minutes before opening.

9. Inlet, minimum pressure and blowdown valves

These valves control loading, net pressure and venting at stop.

Troubleshooting: Inlet valve

Symptom: Compressor runs but does not load; pressure does not rise; no alarm on display.

Possible cause: Inlet (intake) valve stuck closed or its diaphragm torn.

Procedure:

1. Isolate and depressurize.
2. Remove the inlet valve.
3. Inspect the diaphragm and piston, replace the service kit.
4. Check the control air lines to the valve.
5. Reassemble and test loading.

Parts: Inlet valve service kit IV-K3

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar.

Troubleshooting: Minimum pressure valve

Symptom: Unit loads but the net pressure builds very slowly; sump pressure normal while line pressure low; possible oil carryover at start.

Possible cause: Minimum pressure / check valve stuck or its spring broken.

Procedure:

1. Isolate and depressurize.
2. Dismantle the minimum pressure valve.
3. Replace the service kit (spring, piston seals).
4. Reassemble and test pressure build-up.

Parts: MPV service kit MPV-K2

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar.

Troubleshooting: Blowdown valve - alarm E302

Symptom: Start failure alarm; motor struggles or trips when restarting shortly after a stop; sump still pressurized.

Possible cause: Blowdown (vent) valve not venting the sump at stop, so the motor restarts against pressure.

Procedure:

1. Isolate and depressurize manually.
2. Check the blowdown valve and its control line.
3. Replace the blowdown valve service kit.
4. Test: the sump should drop below 1 bar within 60 seconds after stop.

Parts: Blowdown valve kit BV-K1

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar.

10. Solenoid valve and safety valve

Troubleshooting: Loading solenoid valve - alarm E410

Symptom: Solenoid valve fault alarm; compressor stays in unload, never loads.

Possible cause: Loading solenoid valve coil burnt or connector loose.

Procedure:

1. Isolate power.
2. Check the solenoid connector and wiring.
3. Measure coil resistance (expected about 60 ohm for 24 VDC).
4. Replace the coil or complete solenoid valve.
5. Test loading.

Parts: Solenoid coil SV-24VDC

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Electrical work: qualified electrician only; verify zero voltage.

Troubleshooting: Safety valve

Symptom: Loud blowing from the safety valve on the separator tank; pressure near maximum.

Possible cause: Working pressure set too close to the safety valve pressure, or safety valve spring weak.

Procedure:

1. Stop the compressor.
2. Check the unload pressure setpoint is at least 1 bar below the safety valve rating.
3. If settings are correct, replace the safety valve (never adjust or plug it).

Parts: Safety valve SV-14bar (if faulty)

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar. Never block or adjust a safety valve.

11. Sensors and controller power

Sensors: pressure transducer 0-16 bar (4-20 mA); discharge temperature PT1000.

Troubleshooting: Pressure transducer - alarm E501

Symptom: Pressure sensor fault alarm; display shows erratic or impossible pressure readings.

Possible cause: Pressure transducer failed or cable damaged.

Procedure:

1. Isolate and depressurize.
2. Check the transducer cable and connector.
3. Measure the 4-20 mA signal.
4. Replace the transducer.
5. Verify reading against a calibrated gauge.

Parts: Pressure transducer PT-16bar

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Release all stored pressure; confirm the sump gauge reads 0 bar. Electrical work: qualified electrician only; verify zero voltage.

Troubleshooting: Temperature sensor - alarm E502

Symptom: Temperature sensor fault alarm; unit will not start; display shows -99 or open circuit.

Possible cause: Discharge temperature sensor (PT1000) open circuit or connector corroded.

Procedure:

1. Isolate power.
2. Check the sensor connector for corrosion.
3. Measure sensor resistance (about 1100 ohm at 25 C).
4. Replace the sensor.
5. Restart and verify temperature reading.

Parts: Temperature sensor PT1000-TS2

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Electrical work: qualified electrician only; verify zero voltage. Hot surfaces: let the unit cool at least 15 minutes before opening.

Troubleshooting: Controller

Symptom: Controller display blank; no lights; compressor does not respond.

Possible cause: Control transformer fuse blown or controller power supply failed.

Procedure:

1. Isolate power.
2. Check the control circuit fuses.
3. Measure the control transformer output (24 VAC).
4. Replace the fuse or power supply.

Parts: Control fuse 2A, Power supply 24V (if faulty)

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Electrical work: qualified electrician only; verify zero voltage.

12. Main motor and electrical

Electrical work must be carried out by a qualified electrician only.

Troubleshooting: Main motor - alarm E301

Symptom: Main motor overload trip; happens under load, more often on hot days or when supply voltage is low.

Possible cause: Main motor overload relay tripped due to high current (low supply voltage or high discharge pressure setting).

Procedure:

1. Isolate power (lockout/tagout).
2. Measure supply voltage on all three phases.
3. Check the overload relay setting against the motor nameplate.
4. Check the working pressure setpoint is not above rated pressure.
5. Reset and monitor motor current under load.

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Electrical work: qualified electrician only; verify zero voltage.

Troubleshooting: Power supply - alarm E310

Symptom: Phase fault alarm; compressor will not start.

Possible cause: Phase loss or wrong phase sequence on the incoming supply (blown fuse or supply work).

Procedure:

1. Isolate power.
2. Check the incoming fuses.
3. Measure the three phase voltages.
4. If supply work was done, check phase sequence and swap two phases at the incoming terminals if reversed.
5. Restart.

Parts: Fuses (if blown)

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Electrical work: qualified electrician only; verify zero voltage.

Troubleshooting: Star-delta starter - alarm E303

Symptom: Star-delta transition fault; contactors chatter at start; motor stalls when switching to delta.

Possible cause: Worn star-delta contactor or faulty transition timer.

Procedure:

1. Isolate power and lock out.
2. Inspect contactor contacts for pitting.
3. Check the star-delta timer setting (typically 6-8 s).
4. Replace worn contactors.
5. Test start sequence.

Parts: Contactor 3RT-40A

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Electrical work: qualified electrician only; verify zero voltage.

Troubleshooting: Main motor bearings

Symptom: Growling or rumbling noise from the motor; vibration; motor current slightly high; motor housing hot.

Possible cause: Main motor bearing worn or lacking grease.

Procedure:

1. Isolate and lock out.
2. Check bearing noise with a stethoscope and measure vibration.
3. Regrease with the specified grease if within interval.
4. Replace bearings if noise continues (motor workshop).

Parts: Motor bearings 6313-2Z (DE/NDE), Bearing grease

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Electrical work: qualified electrician only; verify zero voltage. Heavy lifting: use a hoist to remove the motor.

13. Pressure settings and the air network

Factory settings: load 8.0 bar, unload 9.0 bar, max 10 bar.

Troubleshooting: Pressure settings

Symptom: Compressor loads and unloads very frequently (short cycling); many motor starts per hour.

Possible cause: Pressure band too narrow or air receiver too small for the demand pattern.

Procedure:

1. Check load/unload setpoints (recommended band at least 0.8 bar).
2. Widen the pressure band on the controller.
3. Check the receiver drain and size.
4. Monitor starts per hour (max 10).

Safety: Do not raise the unload pressure above the rated working pressure.

Troubleshooting: Air distribution network

Symptom: Compressor runs loaded almost all the time; line pressure below setpoint; hissing sounds in the plant.

Possible cause: Air leaks in the distribution pipework, fittings or tools exceed compressor capacity.

Procedure:

1. Walk the network during a quiet period and listen for leaks (or use an ultrasonic detector).
2. Tag each leak.
3. Repair fittings, hoses and couplings.
4. Measure load/unload ratio after repair.

Parts: Fittings and seals as needed

Safety: Isolate and depressurize each pipe section before repair. Wear gloves, safety glasses and hearing protection.

14. Condensate and dryer

The RS-55 includes an integrated refrigerated dryer (pressure dew point +3 C).

Troubleshooting: Condensate drain

Symptom: Water in the compressed air lines; condensate drain never discharges or discharges only air.

Possible cause: Electronic condensate drain valve blocked or its timer/strainer faulty.

Procedure:

1. Isolate the drain valve with its ball valve and depressurize it.
2. Clean the strainer.
3. Test the drain with the test button.
4. Replace the drain valve service kit if it does not open.

Parts: Drain valve service kit DV-K1

Safety: Release all stored pressure; confirm the sump gauge reads 0 bar. Wear gloves, safety glasses and hearing protection.

Troubleshooting: Integrated refrigerated dryer (RS-55 only) - alarm E701

Symptom: Dryer high dew point alarm; water in the air lines downstream of the compressor.

Possible cause: Dryer condenser dirty or dryer drain blocked (RS-55 integrated dryer).

Procedure:

1. Isolate power to the dryer.
2. Clean the dryer condenser fins.
3. Check and clean the dryer condensate drain.
4. Check refrigerant pressures (refrigeration technician only).

Safety: Stop the compressor and apply lockout/tagout on the main isolator. Electrical work: qualified electrician only; verify zero voltage. Refrigerant work: certified technician only.

15. Installation and ventilation

Maximum ambient temperature 40 C. Cooling air flow: RS-37 4.5 m³/s, RS-55 6.0 m³/s.

Troubleshooting: Compressor room ventilation - alarm E101

Symptom: High temperature alarms mainly on hot afternoons; compressor room very hot; cooler clean.

Possible cause: Insufficient ventilation of the compressor room; hot air recirculates to the intake.

Procedure:

1. Measure room temperature near the intake (max 40 C).
2. Check exhaust duct and room fans are working.
3. Open or add ventilation; duct the hot cooling air outside.
4. Monitor discharge temperature.

Safety: Wear gloves, safety glasses and hearing protection.

16. Alarm code table

Code	Meaning	See chapter / page
E001	Emergency stop active	p. 5
E101	High discharge temperature	p. 7, p. 8, p. 9, p. 17
E210	Air filter differential pressure high	p. 6
E220	Separator differential pressure high	p. 10
E301	Main motor overload	p. 14
E302	Start failure	p. 11
E303	Star-delta transition fault	p. 14
E305	Fan motor overload	p. 9
E310	Phase failure / phase sequence	p. 14
E410	Loading solenoid fault	p. 12
E501	Pressure sensor fault	p. 13
E502	Temperature sensor fault	p. 13
E601	Service due	p. 7
E701	Dryer high dew point (RS-55)	p. 16